BME353- COMPUTER AIDED MACHINE DRAWING-I LAB

Course Outcome (CO)		Bloom's Knowledge Level (KL)
	At the end of course , the student will be able to	
CO 1	Understand and apply 2D software to develop a part model	**K3**
CO 2	Understand about temporary and permanent fasteners	**K2**
CO 3	Understand the need for free hand sketching, Free hand sketching of foundation bolts etc.	**K2**
CO 4	Create assembly drawing of simple machine elements like rigid or flexible coupling	**K3**
CO 5	Create 2D drawings and assemblies of various machine components	**K3**
	DETAILED SYLLABUS	**0-0-2**

UNIT-I

Computer aided drafting (1 drawing) Introduction to computer aided drafting; advantages and applications of CAD, concepts of computer aided 2D drafting using any drafting software like AutoCAD, Pro-E, Solid Edge, Draft Sight etc., basic draw and modify commands, making 2D drawings of simple machine parts. Course Outcomes: Upon completion of this course, the students can use computer and CAD software for modelling mechanical components.

UNIT-II

Fasteners (2 drawing sheets) Temporary and permanent fasteners, thread nomenclature and forms, thread series, designation, representation of threads, bolted joints, locking arrangement of nuts, screws, washers, foundation bolts etc., keys, types of keys, cotter and knuckle joints.

UNIT-III

Riveted joints (1 drawing sheet) Introduction, rivets and riveting, types of rivets, types of riveted joints, drawing of boiler joints etc. Free hand sketching (1 drawing sheet) Introduction, need for free hand sketching, Free hand sketching of foundation bolts, studs, pulleys, couplings etc.

UNIT-IV

Assembly drawing (2 drawing sheets) Introduction to assembly drawing, drawing assembly drawing of simple machine elements like rigid or flexible coupling, muff coupling, Plummer block, footstep bearing, bracket etc.

Books and References:

1. Fundamentals of Machine Drawing by Sadhu Singh & Shah, PHI.
2. Engineering Drawing by Bhat, & Panchal, Charotar Publishing House.
3. Machine Drawing with AutoCAD by Pohit and Ghosh, Pearson.
4. Machine Drawing-KL Narayana, P Kannaiah, KV Reddy, New Age.
5. Machine Drawing, N. Siddeshswar, P Kannaiah, VVS Shastry, Tata McGraw Hill.
6. Engineering Drawing, Pathak, Wiley.
7. Textbook of Machine Drawing, K C John, PHI.
8. AutoCAD 2014 for Engineers & Designers, Bhatt, WILEY
9. Engineering Drawing, 2nd edn by Shah & Rana
10. Fundamentals of Engineering Drawing, 11th edn by Warren J. Luzadder, Pearson Education

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